

Linear programming

Background material

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Linear program

Linear program, standard form:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \\ & x \geq 0, \end{aligned}$$

with $x, c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$.

Assumptions: $\text{rank}(A) = m$, $m \leq n$.

We write $A_j \in \mathbb{R}^m$ for the j -th **column** of A , so that the constraints read $\sum_{j=1}^n x_j A_j = b$: a linear program asks for a **non-negative** combination of the columns of A that reproduces b , at minimum cost.

Its feasible set is a **polyhedron**: a finite intersection of **half-spaces** $\{x \mid a^T x \leq \beta\}$, bounded by the **hyperplanes** $\{x \mid a^T x = \beta\}$.

Standard form

Any linear program can be transformed to the standard form. Consider for instance the program

$$\min_x c^T x \text{ s.t. } Ax \geq b.$$

Note that there is no sign constraint on x .

We first add a vector z of surplus variables:

$$\begin{aligned} \min_{x,z} \quad & c^T x \\ \text{s.t.} \quad & Ax - z = b, \\ & z \geq 0. \end{aligned}$$

We next decompose x as the difference of two non-negative variables:

$x = x^+ - x^-$, where $x^+ = \max\{x, 0\} \geq 0$, and $x^- = \max\{-x, 0\} \geq 0$.

Standard form (cont'd)

The problem becomes

$$\begin{aligned} \min_{x^+, x^-, z} \quad & (c^T \quad -c^T \quad 0^T) \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix}, \\ \text{s.t.} \quad & (A \quad -A \quad -I) \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix} = b, \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix} \geq 0. \end{aligned}$$

The inequality constraints of the form $x \leq u$ or $Ax \leq b$ can be handled by adding slack variables:

$$\begin{aligned} x \leq u &\Leftrightarrow x + w = u, \quad w \geq 0, \\ Ax \leq b &\Leftrightarrow Ax + y = b, \quad y \geq 0. \end{aligned}$$

Basic solutions

Without loss of generality, suppose the first m columns of A are independent, and partition A and x accordingly,

$$A = (B \ D), \quad x = \begin{pmatrix} x_B \\ x_D \end{pmatrix}, \quad Bx_B + Dx_D = b,$$

where the **basis matrix** $B \in \mathbb{R}^{m \times m}$ is invertible.

- **Basic solution**: the solution obtained by setting the $n - m$ **non-basic** variables to zero, $x_D = 0$, which leaves $x_B = B^{-1}b$ for the **basic** ones.
- **Feasible basic solution**: a basic solution with $x_B \geq 0$; then $x \geq 0$, and x is feasible.
- **Degenerate basic solution**: a basic solution with some components of x_B equal to zero. More than $n - m$ variables then vanish, and the same point may come from several bases.

A basis of columns, and the coordinates of b

The m columns of B are independent: they form a **vector basis of $\mathbb{R}^m$** . Choosing a basis is choosing m columns of A , and the basic variables are nothing but the **coordinates of b in that basis**,

$$b = \sum_{j \in \mathcal{B}} (x_B)_j A_j, \quad x_B = B^{-1} b,$$

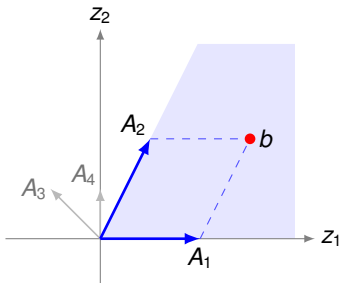
where $\mathcal{B}$ denotes the set of indices of the chosen columns, the $n - m$ non-basic variables being set to zero. The basis is **feasible** exactly when these coordinates are non-negative, i.e. when b lies in the **cone** spanned by the chosen columns,

$$\text{cone}\{v^1, \dots, v^k\} \stackrel{\text{def}}{=} \{\lambda_1 v^1 + \dots + \lambda_k v^k \mid \lambda \geq 0\},$$

and not merely in the subspace they span: the span only asks that b be reachable, the cone asks that it be reachable with **non-negative** weights.

Two bases, two sets of coordinates

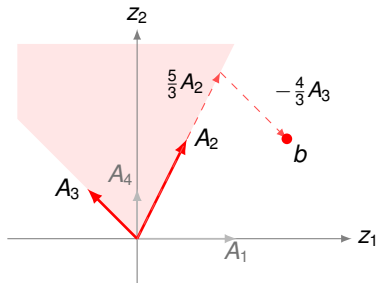
$A = \begin{pmatrix} 2 & 1 & -1 & 0 \\ 0 & 2 & 1 & 1 \end{pmatrix}$, $b = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$: both column pairs below are bases of $\mathbb{R}^2$, only the first one is feasible.



$$B = (A_1 \ A_2), x_B = B^{-1}b = (1, 1) \geq 0$$

$$b = 1 \cdot A_1 + 1 \cdot A_2, x = (1, 1, 0, 0)$$

basic feasible solution



$$B = (A_2 \ A_3), x_B = \left(\frac{5}{3}, -\frac{4}{3}\right) \not\geq 0$$

$$b \notin \text{cone}\{A_2, A_3\}$$

basic, but not feasible

Fundamental theorem of linear programming

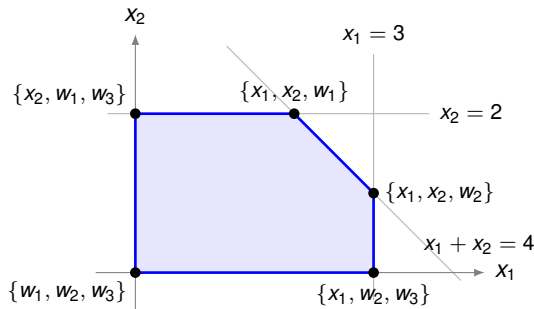
Consider an LP in standard form, with $A \in \mathbb{R}^{m \times n}$, and $\text{rank}(A) = m$.

- If there exists a feasible solution, then there exists a feasible basic solution.
- If there exists a feasible optimal solution, then there exists a feasible optimal basic solution.

The search for an optimum can therefore be restricted to the **finitely many** choices of m columns out of n : this is what the simplex method exploits, moving from one basis to an adjacent one.

Bases and vertices

$\{x \in \mathbb{R}_+^2 \mid x_1 \leq 3, x_2 \leq 2, x_1 + x_2 \leq 4\}$, put in standard form with the slacks w_1, w_2, w_3 : $m = 3$, $n = 5$. Each vertex is labelled by its basis.



The $n - m = 2$ non-basic variables are exactly those vanishing at the vertex: the five vertices are the five feasible bases among the $\binom{5}{3} = 10$ column triples (two of which are singular, three basic but infeasible).

Convex sets and convex functions

- A **convex combination** of $x^1, \dots, x^k$ is $\sum_{i=1}^k \lambda_i x^i$ with $\lambda \geq 0$ and $\sum_{i=1}^k \lambda_i = 1$. Dropping the normalisation $\sum_i \lambda_i = 1$ gives back the non-negative combinations that generate a cone.
- A set C is **convex** if it contains the convex combinations of its points, equivalently the segment $[x, y]$ for all $x, y \in C$. A half-space is convex, and so is any intersection of half-spaces: **a polyhedron is convex**.
- A function f is **convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad x, y \in \mathbb{R}^n, \lambda \in [0, 1].$$

These two notions — convex set, convex function — are really one, as the next slide shows.

Convexity through the epigraph

The **epigraph** of f is the set of points lying above its graph,

$$\text{epi } f \stackrel{\text{def}}{=} \{(x, \mu) \in \mathbb{R}^n \times \mathbb{R} \mid \mu \geq f(x)\},$$

and it turns one notion into the other:

$$f \text{ convex} \iff \text{epi } f \text{ convex.}$$

Everything said about convex **sets** therefore applies to convex **functions**, read one dimension higher — supporting hyperplanes in particular, which is how subgradients arise below.

For a maximum of finitely many linear functions, $f(x) = \max_{i=1, \dots, k} \{(a^i)^T x + \beta_i\}$,

$$\text{epi } f = \bigcap_{i=1}^k \{(x, \mu) \mid \mu \geq (a^i)^T x + \beta_i\}$$

is an intersection of half-spaces, hence a polyhedron: **such an f is convex and piecewise linear** — the form the value function of a linear program will take.

Extreme points and extreme rays

Let $P = \{x \mid Ax = b, x \geq 0\}$ be non-empty.

- $x \in P$ is an **extreme point** (or vertex) of P if it is not a strict convex combination of two points of P :

$$x = \lambda y + (1 - \lambda)z, y, z \in P, \lambda \in (0, 1) \implies y = z = x.$$

(P being convex, testing $\lambda = \frac{1}{2}$ is already enough.)

- The **recession cone** of P is $C = \{d \mid Ad = 0, d \geq 0\}$: the directions along which P is unbounded, $x + \alpha d \in P$ for every $x \in P$ and $\alpha \geq 0$. For $d \in C \setminus \{0\}$, the half-line $\{\alpha d \mid \alpha \geq 0\}$ is a **ray** of P .
- That ray is an **extreme ray** if every decomposition of d inside C stays on the ray:

$$d = d^1 + d^2, d^1, d^2 \in C \implies d^1, d^2 \text{ are non-negative multiples of } d.$$

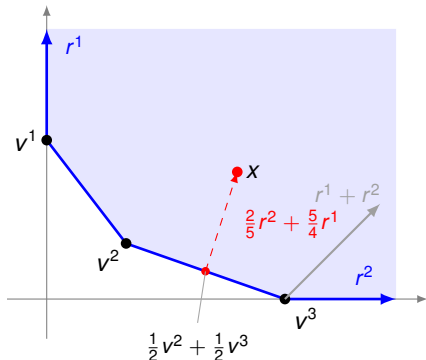
Being half-lines, rays are counted **up to a positive factor**.

Extreme points and extreme rays (cont'd)

- The extreme points of P are exactly its feasible basic solutions. Distinct extreme points have distinct bases, hence there are **finitely many**: at most $\binom{n}{m}$. That bound is crude — most column subsets are singular or infeasible — but the simplex $\{x \geq 0 \mid \sum_j x_j = 1\}$ attains it, with n vertices.
- The rays obey the same mechanism: writing $S = \{j \mid d_j > 0\}$, the ray of $d \in C$ is extreme if and only if the columns $A_j, j \in S$, have rank $|S| - 1$ — they carry exactly one dependency, and d is it. The extreme rays are therefore **finite** in number too.
- More generally, the recession cone of $\{\pi \mid A^T \pi \leq c\}$ is $\{\sigma \mid A^T \sigma \leq 0\}$: this is the form met in the dual problems below.
- **Resolution theorem**: every $x \in P$ is the sum of a convex combination of the extreme points and of a non-negative combination of the extreme rays.

Consequently a linear program over P attains its optimum at an extreme point, or is unbounded along an extreme ray r with $c^T r < 0$.

Extreme points and extreme rays, graphically

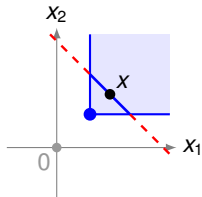


Three extreme points, two extreme rays; $r^1 + r^2$ is a recession direction of P , but not an extreme one, since it splits along two rays of different directions. Every $x \in P$ is a convex combination of the v^i plus a non-negative combination of the r^j . The optimum of $c^T x$ is attained at some v^i , unless $c^T r^j < 0$ for some j .

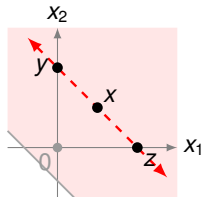
When does a polyhedron have extreme points?

P contains a **line** if $x + \alpha d \in P$ for some $x \in P$, some $d \neq 0$ and *every* $\alpha \in \mathbb{R}$ — equivalently, if d and $-d$ both lie in the recession cone.

A non-empty polyhedron has an extreme point **if and only if** it contains no line. It is then said to be **pointed**.



(a) $P = \{x \mid x \geq (1, 1)\}$: every line leaves P **pointed**, extreme point $(1, 1)$, and $0 \notin P$



(b) $P = \{x \mid x_1 + x_2 \geq -1\}$: the line stays in P $x = \frac{1}{2}(y + z)$: **no extreme point**, though $0 \in \text{int } P$

Translation leaves C unchanged, so the origin plays no role. In standard form $C \subseteq \mathbb{R}_+^n$, and $d, -d \in C$ force $d = 0$: $x \geq 0$ alone makes P pointed. The dual polyhedron is shaped like (b), where $\text{rank}(A) = m$ excludes lines.

Duality: symmetric form

- **Primal** program

$$\begin{aligned} \min_{x \geq 0} \quad & c^T x \\ \text{s.t.} \quad & Ax \geq b \end{aligned} \tag{P}$$

- **Dual** program

$$\begin{aligned} \max_{\pi \geq 0} \quad & \pi^T b \\ \text{s.t.} \quad & A^T \pi \leq c \end{aligned} \tag{D}$$

- $A \in \mathbb{R}^{m \times n}$, $c, x \in \mathbb{R}^n$, $\pi, b \in \mathbb{R}^m$: x gathers the **primal** variables, π the **dual** ones.
- Dualizing (D) gives (P) back: the dual of the dual is the primal.

Duality: from the standard form

The program

$$\begin{array}{ll}\min_x & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array}$$

is equivalent to

$$\begin{array}{ll}\min_x & c^T x \\ \text{s.t.} & Ax \geq b \\ & -Ax \geq -b \\ & x \geq 0\end{array}$$

Duality: asymmetric form

The dual problem can then be written as

$$\begin{aligned} \max_{u,v} \quad & u^T b - v^T b \\ \text{s.t.} \quad & u^T A - v^T A \leq c^T \\ & u \geq 0 \\ & v \geq 0 \end{aligned}$$

or, with $\pi = u - v$,

$$\begin{aligned} \max_{\pi} \quad & \pi^T b \\ \text{s.t.} \quad & \pi^T A \leq c^T \end{aligned}$$

Asymmetric form: $\pi \in \mathbb{R}^m$ is unrestricted in sign.

Primal-dual conversion

Minimization	Maximization
Constraints	Variables
$\geq$	≥ 0
$\leq$	≤ 0
$=$	unconstrained
Variables	Constraints
≥ 0	$\leq$
≤ 0	$\geq$
unconstrained	$=$

Weak duality

(Asymmetric or symmetric form)

If x and π are feasible for the primal and the dual, respectively, then

$$c^T x \geq \pi^T b$$

Proof.

For any primal feasible x and dual feasible π , we have

$$\pi^T b \leq \pi^T Ax \leq c^T x,$$

the first inequality because $\pi \geq 0$ and $Ax \geq b$ (an equality in the asymmetric form, where $Ax = b$), the second because $x \geq 0$ and $A^T \pi \leq c$. □

Corollary

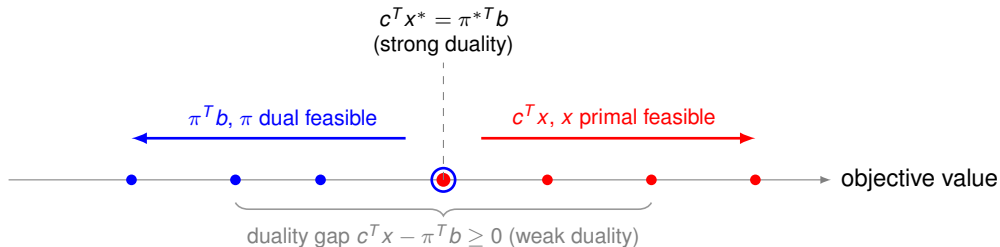
If x_0 and π_0 are feasible for the primal and the dual, respectively, and if

$$c^T x_0 = \pi_0^T b,$$

then x_0 and π_0 are optimal for their respective problem.

But we have said nothing about the feasibility of one problem with respect to the other one!

Weak and strong duality, graphically



Every dual feasible value bounds every primal feasible value from below. Strong duality states that the two families actually **meet**: the best lower bound is the optimal value. The corollary above is read on the picture: a blue and a red mark that coincide can only sit at the meeting point.

Strong duality

If one of the problems (P) or (D) has a finite optimal solution, the other problem also has a finite optimal solution, and the corresponding values of the objective functions are equal. If one of the problems has an unbounded objective, the other problem has no feasible solution.

If a program is infeasible, this does not imply that its dual is unbounded: it can be infeasible as well. In the table below, “bounded” means that the program is feasible and has a finite optimal value.

Primal / Dual	Bounded	Unbounded	Infeasible
Bounded	possible	impossible	impossible
Unbounded	impossible	impossible	possible
Infeasible	impossible	possible	possible

Polyhedral cones and Farkas' lemma

Positive hull: the cone of the previous slides, now generated by **all** the columns of A at once,

$$\text{pos } A = \text{cone}\{A_1, \dots, A_n\} = \{z \mid z = Ax, x \geq 0\}.$$

It is exactly the set of right-hand sides z for which $Ax = z, x \geq 0$, admits a solution.

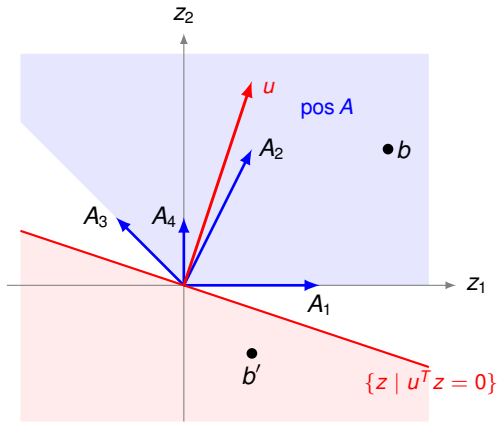
Farkas' lemma

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following holds:

- $\exists x \geq 0$ such that $Ax = b$, i.e. $b \in \text{pos } A$;
- $\exists u \in \mathbb{R}^m$ such that $A^T u \geq 0$ and $b^T u < 0$.

The second vector is a **certificate of infeasibility**: it defines a hyperplane separating b from $\text{pos } A$. One may always take u on an extreme ray of $\{u \mid A^T u \geq 0\}$, so that finitely many certificates suffice.

Farkas' lemma, graphically



Same columns as before: $b = (3, 2)$ lies in the blue cone, so $Ax = b, x \geq 0$ is solvable. $b' = (1, -1)$ does not, and $u = (1, 3)$ certifies it: $A^T u = (2, 7, 2, 3) \geq 0$ while $u^T b' = -2 < 0$. The red half-space $\{z \mid u^T z < 0\}$ holds b' and misses the cone.

The polar cone

The **polar** of a cone $C \subseteq \mathbb{R}^m$ is

$$C^\circ \stackrel{\text{def}}{=} \{u \mid u^T z \leq 0 \ \forall z \in C\},$$

the directions making an obtuse angle with everything in C . For $C = \text{pos } A$ it is read off A :

$$\{u \mid A^T u \leq 0\} = \{u \mid y^T A^T u \leq 0 \ \forall y \geq 0\} = \{u \mid u^T (Ay) \leq 0 \ \forall y \geq 0\} = (\text{pos } A)^\circ,$$

the first equality because $A^T u \leq 0$ componentwise is the same as $y^T (A^T u) \leq 0$ for all $y \geq 0$.

Farkas' lemma says $b \in \text{pos } A$ iff $A^T u \geq 0 \Rightarrow b^T u \geq 0$; replacing u by $-u$, legitimate since it holds for every u , gives $A^T u \leq 0 \Rightarrow b^T u \leq 0$, that is

$$b \in \text{pos } A \iff b^T u \leq 0 \text{ for every } u \in (\text{pos } A)^\circ.$$

So the polar collects the **certificates of infeasibility**: a single u in it with $b^T u > 0$ proves that $Ax = b$, $x \geq 0$ has no solution. In the recourse problems of the following decks, $\text{pos } W$ is the set of right-hand sides the second stage can absorb, and the elements of its polar are exactly what the decomposition methods turn into **feasibility cuts**.

Optimality conditions for a linear program

Specialising the KKT conditions to a linear program, x is optimal **if and only if** there are $\pi \in \mathbb{R}^m$ and $s \in \mathbb{R}^n$ with

$$Ax = b, \quad x \geq 0 \quad (\text{primal feasibility}),$$

$$A^T \pi + s = c, \quad s \geq 0 \quad (\text{dual feasibility}),$$

$$x_i s_i = 0, \quad i = 1, \dots, n \quad (\text{complementary slackness}).$$

- In LP no constraint qualification is needed, and the conditions are **sufficient** as well as necessary — both are specific to the linear (convex, polyhedral) case.
- They give strong duality in one line:
 $c^T x = (A^T \pi + s)^T x = \pi^T Ax + s^T x = \pi^T b$, using $s^T x = 0$ and $Ax = b$.
- Complementary slackness is what lets one read off the optimal y from an optimal π in the recourse problems of the following decks.

The Lagrangian, the general KKT conditions, constraint qualifications and the duality gap are treated in the **KKT background notes**.

The value function of a linear program

Consider the optimal value as a function of the right-hand side,

$$z(b) \stackrel{\text{def}}{=} \min_x \{c^T x \mid Ax = b, x \geq 0\},$$

with $z(b) = +\infty$ when the program is infeasible: its **domain**
 $\text{dom } z = \{b \mid z(b) < +\infty\}$ is exactly $\text{pos } A$.

By duality, $z(b) = \max_{\pi} \{\pi^T b \mid A^T \pi \leq c\}$, where the feasible set **does not depend on b** . Assume it non-empty (otherwise $z \equiv -\infty$ on $\text{pos } A$); $\text{rank}(A) = m$ makes it pointed, so it has extreme points $\pi^1, \dots, \pi^r$ and, for $b \in \text{dom } z$,

$$z(b) = \max_{i=1, \dots, r} (\pi^i)^T b.$$

Subgradients and supporting hyperplanes

A vector g is a **subgradient** of a convex function f at b if it supports f from below everywhere,

$$f(b') \geq f(b) + g^T(b' - b) \quad \text{for all } b',$$

and the set of them is the **subdifferential** $\partial f(b)$.

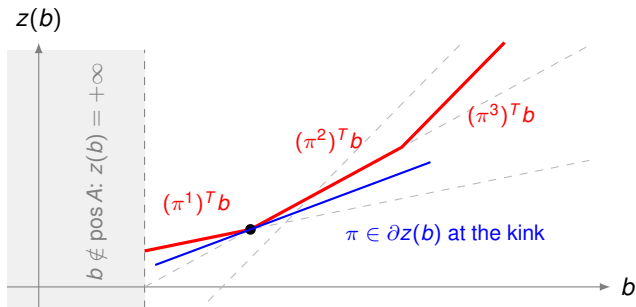
Geometrically, the graph of $b' \mapsto f(b) + g^T(b' - b)$ is a **supporting hyperplane** of $\text{epi } f$ at $(b, f(b))$: it touches f there and stays below it elsewhere. Subgradients are the epigraph reading of convexity, taken at one point.

- $\partial f(b)$ is **convex and closed**: its definition is one linear inequality in g per b' .
- It is **non-empty** at every b in the relative interior of $\text{dom } f$ — there, a convex function never lacks a supporting hyperplane. A cut can thus be generated at **every** iterate.
- For a differentiable f , $\partial f(b) = \{\nabla f(b)\}$; at a kink the subgradient is not unique.

The value function: properties

- As a maximum of finitely many linear functions, z is **convex and piecewise linear** in b .
- Any optimal dual solution π^* at b is a **subgradient**: for any b' , $z(b') \geq (\pi^*)^T b' = z(b) + (\pi^*)^T (b' - b)$. The optimality cuts of the decomposition methods are exactly these supporting hyperplanes.
- On $\text{dom } z$, z is **Lipschitz**, of constant $\max_i \|\pi^i\|$.
- An affine change of right-hand side preserves all of this: $x \mapsto z(h - Tx)$ is again convex and piecewise linear — this is the **recourse function** of the following decks.
- In the costs c , z is a minimum of linear functions: **concave and piecewise linear**.

The value function, graphically



One linear piece per optimal dual vertex. At a kink two bases are optimal and the subgradient is not unique — the blue supporting line is one choice among many; elsewhere the slope is the unique optimal dual solution π^* . This is the function that recourse models differentiate, with $b = h(\xi) - T(\xi)x$.

The subdifferential of the value function

That picture is exact. Let $I(b) = \{i \mid (\pi^i)^T b = z(b)\}$ collect the dual vertices attaining the maximum at b . Then, for b in the interior of $\text{dom } z$,

$$\partial z(b) = \text{conv}\{\pi^i \mid i \in I(b)\}.$$

- $|I(b)| = 1$: z is differentiable at b , and $\nabla z(b) = \pi^*$ is the unique optimal dual solution.
- $|I(b)| > 1$: b is a kink, and $\partial z(b)$ is the polytope spanned by the optimal dual vertices — the slope of the blue line opposite is one of its points.

The extreme points of the dual polyhedron, met at the start of these reminders, are thus exactly the **generators of the subdifferential** of the value function: the polyhedral theory and the convex analysis are two readings of one object. A decomposition method that returns an optimal dual solution returns a vertex of $\partial z(b)$.

References

- D. Bertsimas and J.N. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, 1997 (Chapters 2–4).
- V. Chvátal, *Linear Programming*, Freeman, 1983.
- A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986 (Chapters 7–8, for Farkas' lemma and polyhedral theory).
- J.R. Birge and F. Louveaux, *Introduction to Stochastic Programming*, 2nd edition, Springer, 2011 (Chapter 3, for the value function and the recourse problem).

The Lagrangian, the general KKT conditions and the duality gap are covered in the companion **KKT background notes**.